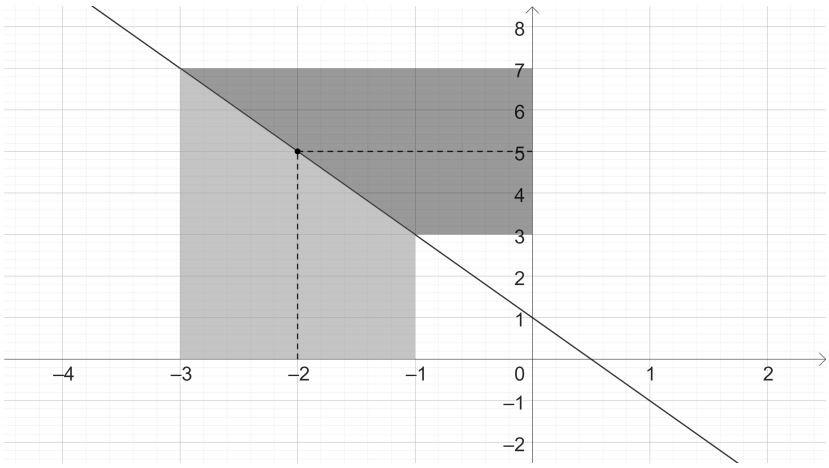
MATH 2413 - Prior Test (SP24)
Brendan Caseria
Test 1

Directions:
- Turn off your cell phone **now**.
- Calculators ARE allowed during the test.
- No notes or external scratch paper allowed. Scratch paper is provided in the back; you may rip it out if you wish, but you must submit it with the test.
- Read all of the directions for each question.
- Answer the questions to the best of your ability in the space provided.
- Show **all** of your work unless otherwise noted.
- **Box** your answers.

Question:	1	2	3	4	5	6	Total
Points:	18	16	16	16	18	16	100
Score:							

1. (a) (6 points) The graph below is a partially labeled diagram depicting an $\varepsilon - \delta$ limit of a linear function. Write out the limit in the form $\lim_{x \to c} f(x) = L$ with the correct c, $f(x)$, and L in place. Also, determine the values of ε and δ shown in the graph. *No need to show work for this part.*



(b) (6 points) For this limit, find δ in terms of ε generally. *Note: we are no longer looking at the specific values of ε and δ from part a). This is the "scratch work" part of a $\varepsilon - \delta$ proof.*

(c) (6 points) Write a formal proof for the limit. Start by filling in the blanks, then proceeding using algebra.

Let $\varepsilon >$ _____ and $\delta =$ _____. Then, we have . . .

2. Compute the following limits. Note: You may not use any limit rules that involve derivatives for this problem (i.e. L'Hopital's Rule).

   (a) (4 points) $\lim_{x \to 2} \frac{\log_2(x)}{x-1}$

   (b) (6 points) $\lim_{x \to 3} \frac{x^3 - 9x}{x^2 - 3x}$

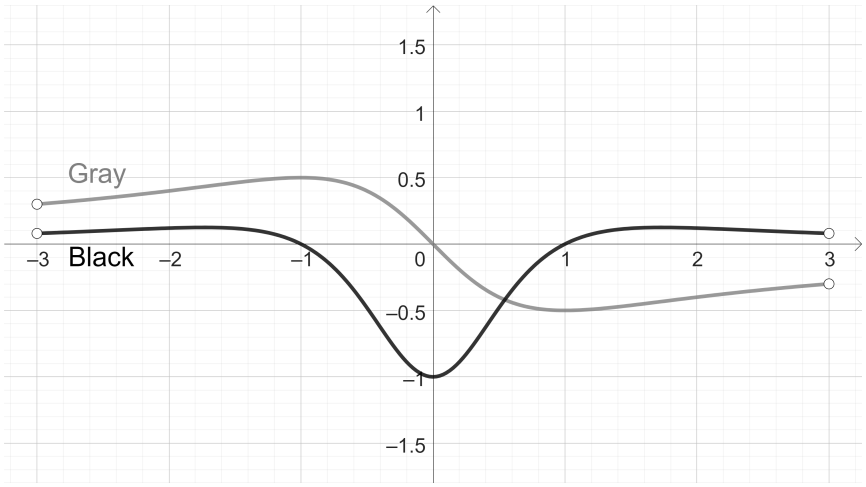
   (c) (6 points) $\lim_{x \to 1^+} \frac{\frac{1}{x} - 1}{x - 1}$

3. Consider the function $f(x) = \dfrac{-x}{x^2 - 1}$.

   (a) (8 points) Find all horizontal asymptotes of $f(x)$. Support your answers with limit statements.

   (b) (8 points) Find all vertical asymptotes of $f(x)$. Support your answers with limit statements.

4. Below are two curves: "Gray" and "Black". Compute the following.



(a) (2 points) The "Gray" function evaluated at $x = 1$. *No need to show work.*

(b) (2 points) The "Black" function evaluated at $x = 1$. *No need to show work.*

(c) (3 points) The slope of the tangent to the "Gray" curve at $x = 1$. Draw this tangent line on the graph above.

(d) (3 points) The slope of the tangent to the "Black" curve at $x = 1$. Draw this tangent line on the graph above. *An estimate is okay here but give a reasonable answer.*

(e) (6 points) One of these curves is the derivative graph of the other curve. In other words, one of these curves is $f(x)$ while the other curve is $f'(x)$. Which one is which? Explain in words how you know. *Hint: Your answers to the previous parts may help here.*

5. Consider the following function, where m and b are unknown constants:

$$f(x) = \begin{cases} \sin(x), & x \leq 0 \\ mx + b, & 0 < x < \pi \\ \cos(x), & x \geq \pi \end{cases}$$

(a) (6 points) Explain in words why this function is definitely continuous for all $x \neq 0$ and $x \neq \pi$ no matter the choice of m and b.

(b) (12 points) Find the values of m and b that make this function continuous everywhere.

6. Consider the function $f(x) = a\sqrt{x}$, where a is some unknown constant.

   (a) (12 points) **Using the limit definition for the derivative**, find $f'(1)$. Work done with derivative rules will not be graded.

   (b) (4 points) Find any value(s) of a that make the tangent line to the curve at $x = 1$ parallel to $y = x$.

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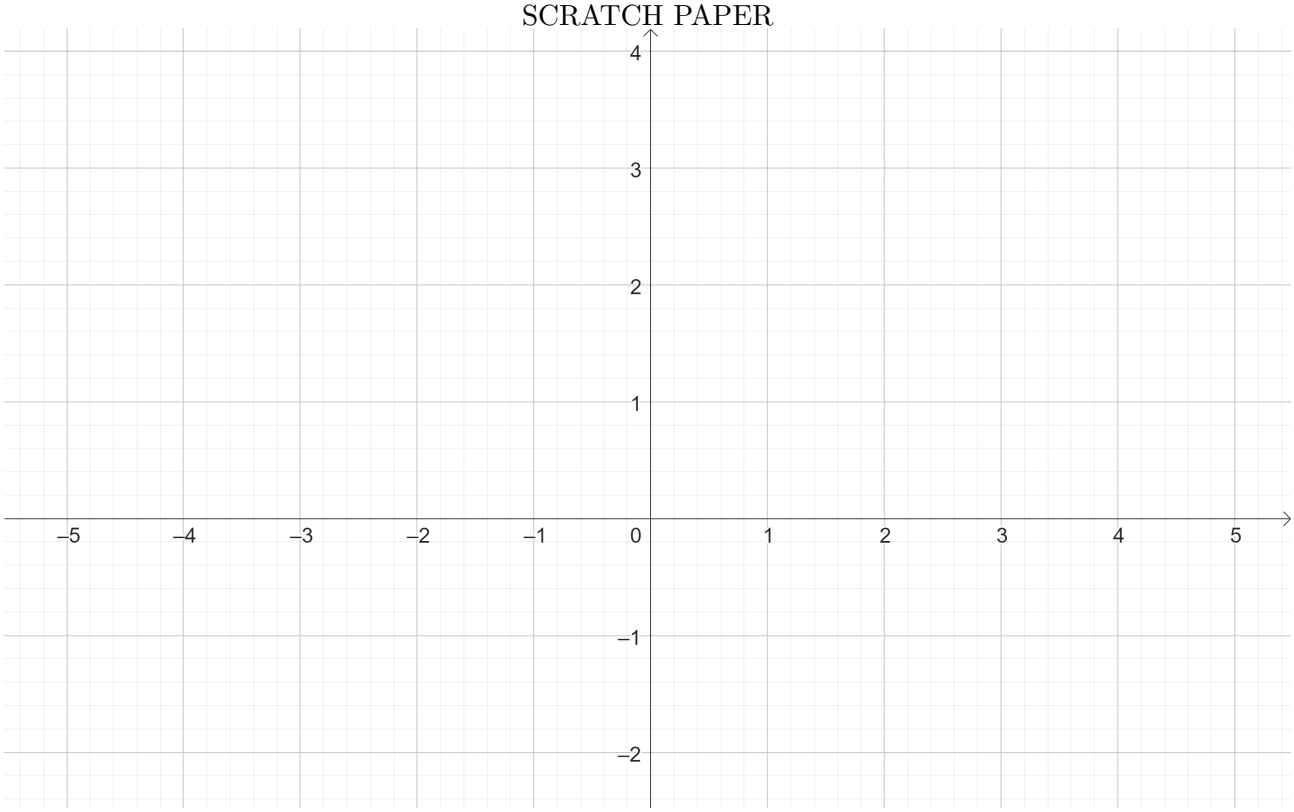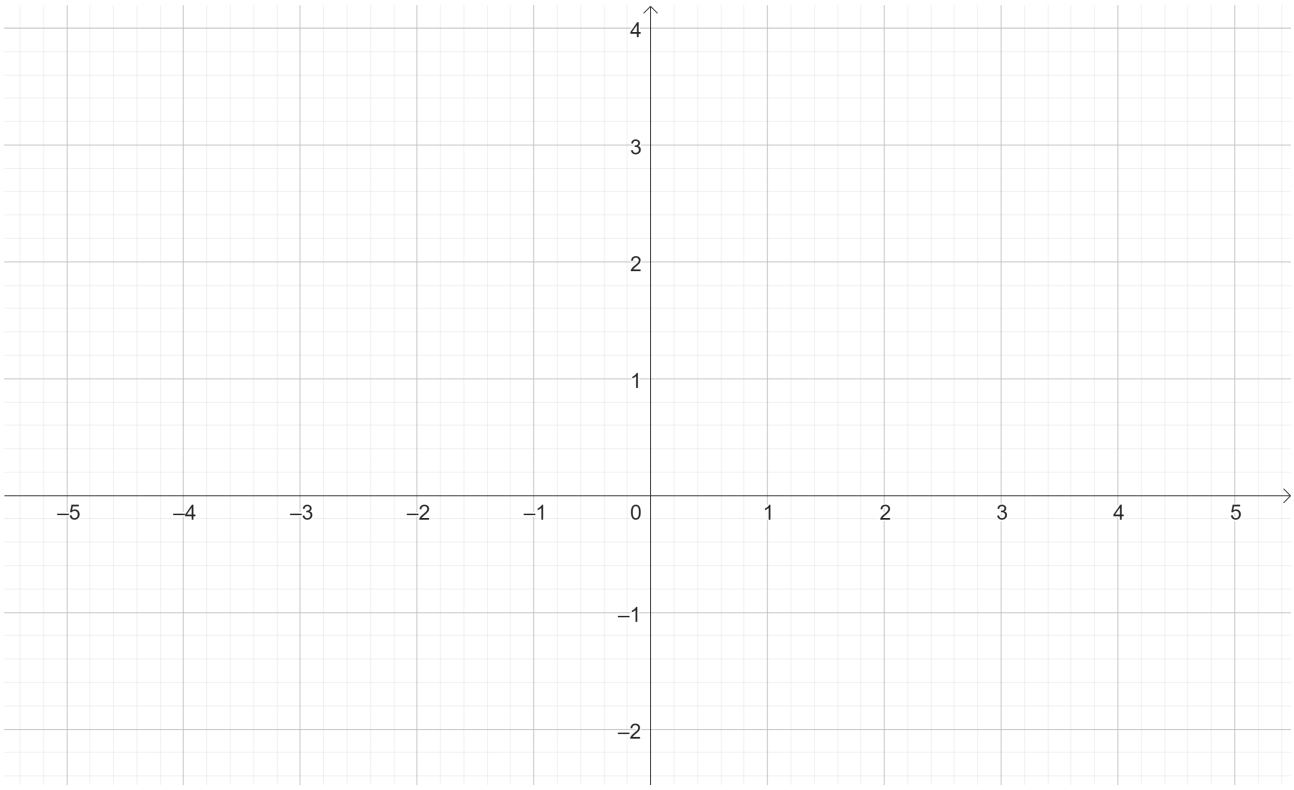
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